

Systems of linear equations

- ▶ set (or *system*) of m linear equations in n variables $x_1, \dots, x_n$:

$$\begin{aligned}A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= b_1 \\A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= b_2 \\&\vdots \\A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= b_m\end{aligned}$$

- ▶ n -vector x is called the variable or unknowns
- ▶ A_{ij} are the *coefficients*; A is the coefficient matrix
- ▶ b is called the *right-hand side*
- ▶ can express very compactly as $Ax = b$

Systems of linear equations

- ▶ systems of linear equations classified as
 - under-determined if $m < n$ (A wide)
 - square if $m = n$ (A square)
 - over-determined if $m > n$ (A tall)
- ▶ x is called a *solution* if $Ax = b$
- ▶ depending on A and b , there can be
 - no solution
 - one solution
 - many solutions
- ▶ we'll see how to solve linear equations later

Outline

Linear functions

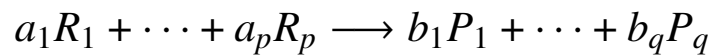
Linear function models

Linear equations

Balancing chemical equations

Chemical equations

- ▶ a chemical reaction involves p reactants, q products (molecules)
- ▶ expressed as



- ▶ $R_1, \dots, R_p$ are reactants
- ▶ $P_1, \dots, P_q$ are products
- ▶ $a_1, \dots, a_p, b_1, \dots, b_q$ are positive coefficients
- ▶ coefficients usually integers, but can be scaled
 - e.g., multiplying all coefficients by $1/2$ doesn't change the reaction

10. Matrix multiplication

Outline

Matrix multiplication

Composition of linear functions

Matrix powers

QR factorization

Matrix multiplication

- ▶ can multiply $m \times p$ matrix A and $p \times n$ matrix B to get $C = AB$:

$$C_{ij} = \sum_{k=1}^p A_{ik}B_{kj} = A_{i1}B_{1j} + \cdots + A_{ip}B_{pj}$$

for $i = 1, \dots, m, j = 1, \dots, n$

- ▶ to get C_{ij} : move along i th row of A , j th column of B
- ▶ example:

$$\begin{bmatrix} -1.5 & 3 & 2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & -2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3.5 & -4.5 \\ -1 & 1 \end{bmatrix}$$

Special cases of matrix multiplication

- ▶ scalar-vector product (with scalar on right!) $x\alpha$
- ▶ inner product $a^T b$
- ▶ matrix-vector multiplication Ax
- ▶ *outer product* of m -vector a and n -vector b

$$ab^T = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & & \vdots \\ a_m b_1 & a_m b_2 & \cdots & a_m b_n \end{bmatrix}$$

Properties

- ▶ $(AB)C = A(BC)$, so both can be written ABC
- ▶ $A(B + C) = AB + AC$
- ▶ $(AB)^T = B^T A^T$
- ▶ $AI = A$ and $IA = A$
- ▶ $AB = BA$ *does not hold in general*